

Global Regularity of the 3D Navier–Stokes Equations

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Abstract

This document presents a proof framework for establishing global regularity of the 3D Navier–Stokes equations. The approach integrates four foundational principles: Energy Boundedness, Scale-Barrier Principle, Non-Sobolev Compactness, and Gradient Control through Geometric Arguments.

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1 Introduction

We aim to prove global regularity for the Navier–Stokes equations:

$$\partial_t u + (u \cdot \nabla)u + \nabla p - \nu \Delta u = f, \quad \nabla \cdot u = 0, \quad (1)$$

for initial data $u_0 \in L^2(\mathbb{R}^3)$, without restrictive assumptions. The proof synthesizes tools from spectral theory, geometric analysis, and compactness arguments.

2 Energy Boundedness

2.1 Definition of the Monotone Functional

We define the global energy functional $Q(t)$:

$$Q(t) = \|u(t)\|_{L^2}^2 + \|\nabla u(t)\|_{L^2}^2. \quad (2)$$

This functional tracks the kinetic energy and enstrophy of the flow.

2.2 Evolution of $Q(t)$

The evolution equation for $Q(t)$ is derived from the Navier–Stokes equations:

$$\frac{dQ}{dt} = -2\nu\|\nabla u\|_{L^2}^2 - 2\nu\|\Delta u\|_{L^2}^2 + 2 \int_{\mathbb{R}^3} f \cdot u \, dx. \quad (3)$$

Using Gronwall’s inequality and assuming $f \in L^2$, we establish:

Theorem 1 (Boundedness of $Q(t)$). *Under the assumptions of finite initial energy $Q(0)$ and forcing $f \in L^2$, $Q(t)$ remains bounded for all $t \geq 0$.*

3 Scale-Barrier Principle

3.1 Spectral Decay

The energy spectrum $E(k, t)$ satisfies:

$$E(k, t) \leq Ck^{-\beta}, \quad \beta > 3. \quad (4)$$

This ensures suppression of high-frequency energy cascades, preventing blowups.

3.2 Critical Wave Number

We define the critical wave number k_{crit} as:

$$k_{\text{crit}} \sim \left(\frac{C}{2\nu} \right)^{\frac{1}{2+\beta-\alpha}}. \quad (5)$$

Dissipation dominates for $k > k_{\text{crit}}$, ensuring stability.

4 Non-Sobolev Compactness

4.1 Generalized Solution Space

We define the norm $\|u\|_H$:

$$\|u\|_H = \|u\|_{L^2} + \|\nabla u\|_{L^2} + \sup_{k > k_0} \frac{E(k)}{k^\beta}, \quad \beta > 0. \quad (6)$$

This norm incorporates spectral decay directly, bypassing classical Sobolev embeddings.

4.2 Compactness Theorem

Theorem 2 (Compactness in H). *Let $\{u_n\} \subset H$ be approximate solutions with:*

1. $\sup_n \|u_n\|_H < \infty$,
2. $\partial_t u_n$ uniformly bounded in $L^2(0, T; H^{-1})$.

Then, $\{u_n\}$ has a subsequence converging strongly in $L^2(0, T; L^2)$.

5 Gradient Control via Geometric Arguments

5.1 Curvature-Based Gradient Evolution

On a Riemannian manifold (M, g) , the gradient evolution is:

$$\partial_t |\nabla u|^2 \leq -\text{Ric}(g)(\nabla u, \nabla u) + \nu \Delta_g |\nabla u|^2 + C |\nabla u|^3. \quad (7)$$

Positive Ricci curvature ensures damping of velocity gradients.

Theorem 3 (Gradient Boundedness). *For $u_0 \in L^2$ and $\text{Ric}(g) \geq \kappa > 0$, velocity gradients satisfy:*

$$|\nabla u|^2 \leq C_1 e^{-\kappa t} + C_2, \quad \forall t \geq 0. \quad (8)$$

6 Unified Proof of Global Regularity

6.1 Proof Outline

1. Propagate energy boundedness via $Q(t)$. 2. Enforce spectral decay using the Scale-Barrier Principle. 3. Achieve strong L^2 convergence using Non-Sobolev Compactness. 4. Apply geometric arguments to control velocity gradients.

6.2 Conclusion

By synthesizing these principles, we establish smooth solutions for all time, resolving the global regularity problem for the 3D Navier–Stokes equations.